

Using Keystroke Dynamics Behavioral Biometrics to Identify Users

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ABSTRACT

This study explores the use of keystroke dynamics as a behavioral biometric for user identification. Unlike physiological biometrics, such as fingerprints or facial recognition, keystroke dynamics leverages the unique typing patterns of individuals to create a distinctive signature. This goal of this research was to develop a machine learning-based system that utilizes keystroke dynamics for continuous and unobtrusive user authentication. By collecting and analyzing keystroke data from multiple users, relevant features were extracted and used to train a machine learning model to identify users with high accuracy. This study shows that using keystroke dynamics for behavioral biometrics is possible for creating a scalable authentication system that can provide an additional layer of security on top of traditional security measures.

CCS CONCEPTS

• **Computing methodologies** → **Supervised learning by classification**; **Neural networks**; • **Security and privacy** → **Biometrics**; • **Human-centered computing** → *Keyboards*.

KEYWORDS

Behavioral Biometrics, Machine Learning, Security, Authentication, Keyboard Behavior, Siamese Network, Feature Extraction

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1 INTRODUCTION

Biometrics is a type of authentication that is commonplace in the current technological landscape. There are two primary types of biometrics, physiological and behavioral. Physiological biometrics are things like fingerprints, facial recognition, and other static information about a person. These types of biometrics are extremely common in practical applications. The less common category is behavioral biometrics, which identifies people by their dynamic actions such as how they interact with devices, how they speak or how they move [7].

One of these Behavioral Biometrics is Keystroke Dynamics. Every person has a unique way they interact with the keyboard, which in turn creates a unique signature that can be used to identify them [7]. Many different behavioral biometrics can be used for identification, such as screen touches on mobile devices, mouse movement, keystroke dynamics, voice, gait, and various other behaviors, as

well as combinations of these measures. The objective of this study is to create a biometrics system that uses only keystroke dynamics to authenticate users.

Many similar studies focus on keystroke dynamics in combination with other data sources. Like on mobile devices where the keystroke dynamics are used alongside things like motion sensor readings and finger movements [10, 12]. Another study used a camera pointed at the keyboard to study hand movement patterns rather than keystroke dynamics [1]. This information is not available on all devices like computers, however.

Using purely keystroke dynamics is advantageous due to its accessibility. Almost anyone interacting with a computer will have to use the keyboard for some tasks. It is also unobtrusive, as keystroke dynamics information can be collected while doing ordinary tasks on a device, and the user does not need to go out of their way to perform specific actions like using their voice, using the mouse, or doing something within view of a camera. Keystroke dynamics can also be continuous, allowing for authentication throughout sensitive processes, rather than only at the beginning.

With these considerations in mind, this study aims to implement a behavioral biometrics system with machine learning using only keystroke dynamics. This method should be unobtrusive for the user while adding an extra layer of security. Another primary goal of the study is to have this authentication system be continuous. That means having authentication that can run in the background over the course of a process that is constantly checking the users' biometrics. Lastly, this model should ideally be scalable. If the model can easily adjust to new user biometrics, it will be more flexible to implement in real-world applications.

To ensure that these goals are met, the final model should achieve an Equal Error Rate (EER) of 5% or below, with greater than 90% accuracy. Meeting these metrics will ensure ease of use for the authentication system and indicate a successful result. This is not as strict as it would be for a primary authentication system, but keyboard authentication would likely be a tertiary security measure in practice.

2 BACKGROUND RESEARCH

Behavioral biometrics has an interesting history, with the identification of unique typing patterns going back all the way to WWII. "The first mention of the use of the typing method to identify an individual can be dated to the period of the Second World War. Back then, a common way of communication was to send messages using the Morse code, and operators quickly learned how to write correspondence" [3]. Since the invention of machine learning, various behavioral biometrics have been studied for authentication purposes.

One study [2] was done that simulated an office environment for 80 workers where information like keystroke dynamics, mouse dynamics, browsing history, and applications used were collected. Users also had to take numerous psychometric tests where further data was collected. Features were extracted from this data and used to train models to identify individual users with reasonable levels of accuracy.

Later a study [1] took a different approach, using computer vision with a camera pointed at the keyboard to identify users. This method extracts features from the video stream and uses a regression model to compare to a saved “probe”. This method shows promise with a very high accuracy result, however implementing such a system practically has the obvious drawback of requiring a camera set up above a keyboard implementing this system.

More recently a method was proposed using a similar feature subset to this study and gives an outline for how such a keystroke authentication system could be implemented [7]. With mobile device security becoming more important, behavioral biometrics was studied for use on mobile devices using both keystroke dynamics as well as other device sensors like tilt and swipe patterns [8]. This is similar to studies [9, 10, 12] which used similar features on a mobile device to classify users.

Some common models used are RNNs as used in [11], LSTMs, SVMs as used in [4], Decision Trees, and Random Forests. The proposed model for this study, however, is a Siamese Network. Siamese networks were first proposed in 1993 but have found significant use in recent years when combined with more powerful networks and deep learning [5]. The most similar research to this study is [6], where a Siamese Network was used to identify users based on keystroke dynamics when entering a password with an 88% accuracy. The aims of this study are to create a model that is more broadly applicable to any typing situation. With additional features as well as novel model architecture, a more generally applicable and accurate model can be created, furthering the research on this approach.

3 METHODOLOGY

3.1 Method Outline

For this study, a machine learning model was created that can identify users using their unique keyboard dynamics. This model was built using Python TensorFlow and Keras. The model was designed with scalability in mind, which means it needs to be highly adaptable to different typing patterns and not specifically tuned to typing a specific phrase or password. The features extracted from the data were selected with this in mind and should be applicable in most typing situations across a wide range of durations. To facilitate this model, a Siamese Network (SSN) was used to compare stored profiles to target probes.

3.2 Data Collection

The first step of the process is to collect data from multiple different users. For this, a web based front-end interface was created to collect keyboard data and store results in a database. The application was then given to multiple individuals who were asked to complete a typing task for 60 seconds while keyboard data is collected. In total, data was collected on 15 different users. This data collected includes

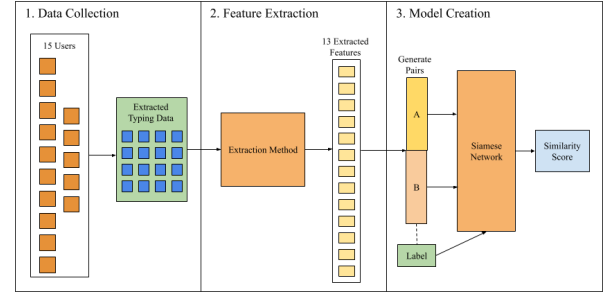


Figure 1: Methodology phases outline.

the key that was pressed, the timestamp when it was pressed, and the timestamp when it was released. This keyboard data is used for the next step, feature extraction.

3.3 Feature Extraction

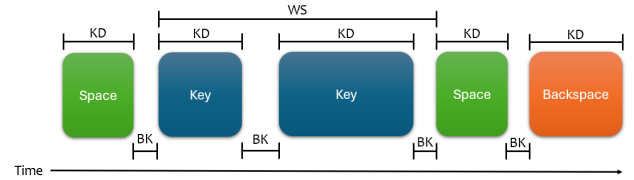


Figure 2: The features extracted based on the timing results of various keyboard actions.

A total of 13 features were extracted from the data to get a unique profile of each typist. The mean, median, range, and standard deviation were collected on the following:

- Time between each key press (BK)
- The ratio of the time taken to write a word to its length (WS)
- The time each key was down (KD)

Additionally, the ratio of backspace keys to total keys pressed was measured. These features are common across several studies like [6, 7]. WS and backspace ratio were additions made for this study. These features were selected because they provide useful information to differentiate between users while also not restricting the possible typing scenarios too much by broadening the required data collection window.

Another step performed after feature extraction is normalization. Normalization was done using Z-Score Normalization.

$$Z = \frac{x - \mu}{\sigma}$$

- Z - the standardized feature
- x - the original feature
- μ - the average for that feature
- σ - standard deviation

Normalizing the features ensures that training can be more consistent when used as an input to a machine learning model, especially ones like an SSN that rely on distance calculations.

Feature vectors were extracted on the full 60 second dataset for each user to represent their profile or signature. To gain data representing dynamic user typing, data was extracted on individual 20 second splits from their overall dataset. To combine this data into usable examples for the machine learning model. Pairs were created with every user's signature paired with data from every user's 20 second splits (probe).

$$Pair = [F_{signature}, F_{probe}]$$

The label is set to 1 if $F_{signature}$ and F_{probe} come from matching users, and 0 if they are from different users.

3.4 Model Creation

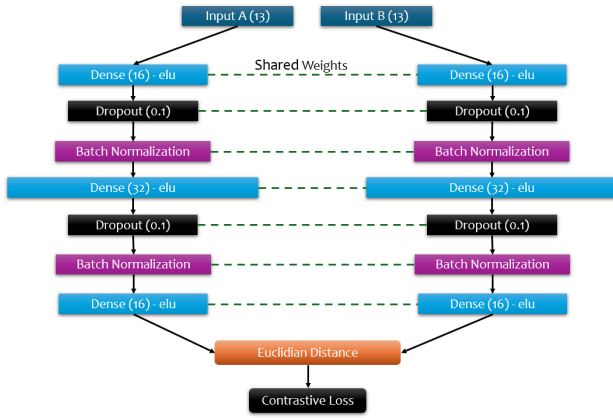


Figure 3: Siamese network architecture.

Many of the past solutions implemented for keystroke dynamics behavioral biometrics were trained to either identify a single user in a binary classification task, or a trained set of users. This requires additional training with each person added. Also, training these models requires a subset of data from multiple other people which makes it difficult if you are starting with a small set of individuals. For scalability reasons, having to re-train the model or set of models with every new user is not ideal for a production environment. That is why, for this study, a Siamese Network is used to learn a loss function on the data which can be generalized to new predictions. The structure of this network, shown in fig. 2, uses two identical Deep Neural Networks (DNNs). Input A is the saved profile (signature) of a user which is a normalized feature vector. Input B is the target feature vector that the saved profile will be compared against. Each DNN will output a feature vector:

$$InputA \rightarrow [F_1], InputB \rightarrow [F_2]$$

Euclidean distance is then calculated on these features with D dimensions:

$$L2(F_1, F_2) = \sqrt{\sum_{i=0}^D (F_{2i} - F_{1i})^2}$$

Contrastive loss is used on the L2 distance to optimize the output distance to be larger when profiles come from different users and

smaller when profiles come from the same user. Contrastive loss is measured on N outputs from the current batch:

$$Contrastive Loss(y_{true}, y_{pred}, m) = \frac{1}{N} \sum_{i=0}^N [y_{true}^i \cdot (y_{pred}^i)^2 + (1 - y_{true}^i) \cdot \max((m - y_{pred}^i), 0)^2]$$

The value of y_{true}^i is either 1 or 0 based on whether there is a match or not. y_{pred}^i is the L2 distance calculated on the input feature vectors. m is a margin value that was set to 1 for this study. If there is a match between the signature and the probe, the first term is in play. $(y_{pred}^i)^2$ This will punish the network for having a large distance between feature vectors from the same user. If the label is 0, the second term is in play. $\max((m - y_{pred}^i), 0)^2$ This will punish the network for having a small distance between non-matching users. As a result, each DNN sub-network will learn weights that make distances for matching inputs very small, and distances for non-matching inputs very large.

To ensure the model is scalable. Measures were taken to prevent overfitting on the data. Dropout and Batch Normalization layers are used in conjunction with early stopping on a 20% validation split when training to reduce overfitting. Due to the small dataset, there are a lot more examples of the negative case (users not matching) than the positive (users matching), to prevent bias while training, random oversampling was used to duplicate cases of the positive class to match the negative. After oversampling, the training set was an even split across 1476 examples.

The model was trained with a batch size of 64 over a maximum of 200 epochs, halted by early stopping.

4 EXPERIMENTS

Various experiments with different models and hyperparameters were done during the development process that helped narrow down the methodology.

4.1 Experiment 1

A non-machine learning method was tried using normalized signatures that were compared with cosine similarity. This method showed some level of correlation between matching signatures, but the false positive and negative rate was very high, and it became clear that a machine learning method would be needed to learn more complex relationships within the data.

4.2 Experiment 2

Larger and more layers of dense networks were tested for the individual DNNs, however this seemed to make no difference in quality while slowing down processing significantly. Therefore, a smaller network was chosen.

4.3 Experiment 3

Cosine similarity was tested as a distance function in place of Euclidean distance, however Euclidean distance was by far the best performing.

4.4 Experiment 4

Various hyperparameters were tuned for better results. Initially higher dropout rates (0.3-0.5) were used, but that seemed to hurt performance too much with the smaller size of the dense layers, so a smaller dropout rate of 0.1 was chosen. Various activation functions were also tested like ReLU, Leaky-ReLU, and ELU. ELU had the best performance and was chosen as the activation function.

5 RESULTS AND ANALYSIS

For testing the final model, the standard accuracy metric was used in addition to the false acceptance rate (FAR) and false rejection rate (FRR). The equal error rate (EER) is calculated from the FAR and FRR. All these metrics vary based on the threshold (θ) chosen.

$$FAR = 100 \cdot \frac{\text{False acceptances} | \theta}{\text{Total Fraudulent Attempts}}$$

$$FRR = 100 \cdot \frac{\text{False rejections} | \theta}{\text{Total Legitimate Attempts}}$$

$$EER = FAR \cap FRR$$

To find the optimal threshold (θ), FAR and FRR are calculated on the range $\theta : 0 \rightarrow 1$ and θ is set at the point of the EER. A lower equal error rate means both better security and less intrusiveness on the user side with a lower rate of both false acceptances and rejections. For this study, a target EER of 5% was chosen. While this is on the higher side for important security operations, keystroke behavioral biometrics is meant to be an additional layer of security, not the primary source, so a higher target EER is an acceptable goal.

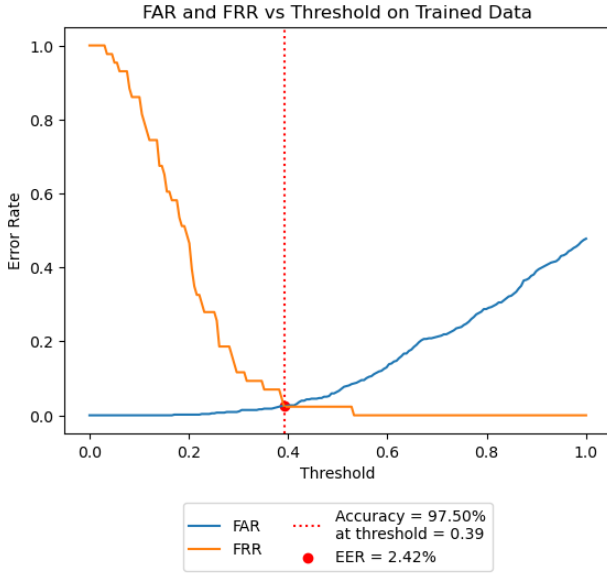


Figure 4: Accuracy FAR, FRR, and EER at best threshold on trained data.

Figure 4 indicates the EER, and accuracy of the final model on the training set. The threshold selected (0.39), is the optimal position to minimize the FAR and FRR with a resulting EER of 2.42%. This

means, of the negative examples, only 2.4% were able to gain false access to the system, and only 2.4% of positive examples were rejected. This is below the EER target of 5% and indicates a robust system. The overall accuracy of this model on the testing data was 97.5%.

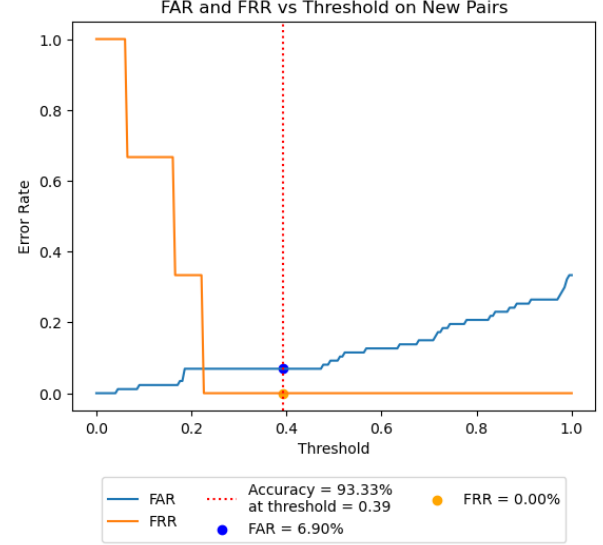


Figure 5: Accuracy FAR, and FRR, at best threshold on new data.

Figure 5 shows the same threshold but used on data that the model hasn't seen before. This means new user profiles and new positive and negative examples. At the optimal threshold, a false acceptance rate of 6.9% was measured with a 0% false rejection rate. Overall, the model performed with 93.33% accuracy on new data. These results outperform the primary comparison for this study [6] which achieved an accuracy of 89% and EER of 27% on new data, however the window for data collected in this study is much higher than a small password window which may account for some of the differences.

These results show some falloff for new data, but it is not significant. The model still performs with a greater than 90% accuracy even on new data which is within the goals for the study. Though the FAR is slightly above the target goal of 5%, which indicates some falloff in quality.

6 CONCLUSION

6.1 Findings

The results from this study show that keystroke behavioral biometrics by themselves are enough to accurately identify users. Furthermore, using a Siamese Network to differentiate user signatures on a generalized feature set shows high effectiveness and scalability, with similar quality results on new data without the need for retraining.

6.2 Future Study

Further study could be done to optimize the model for a production environment. More testing would need to be done on scalability with a much larger set of users as the data set collected for this study was small. Extracting additional features or combining with other biometric types, such as mouse dynamics, could improve model performance at scale while remaining unobtrusive.

There are some practical issues that still need to be solved before a system like this can be implemented in a production environment. One potential issue is people's typing behavior changing over time. They could be using a new keyboard, have an injury, be getting more skilled at typing, or just be tired that day. For more gradual changes, the saved profile could be updated over time with more recent data, keeping the signature close to a user's current typing style. More drastic changes like an injury may require generating a new signature. Different signatures could also be stored and used based on which device, and therefore keyboard, a user is currently using.

Privacy concerns are also a relevant issue when implementing a system like this. Users would need to be made aware of the data being collected on them, and any implementation would have to conform to relevant privacy laws.

The implementation of this system would also need to remain out of the way for normal users. False rejections have a high cost if it means locking the user out of whatever they were doing or causing annoyance. One possible way to reduce this issue is explored in the study by Sağbaş where a circular queue was used, and a user was only rejected if all checks within the queue indicated the user was invalid [9]. The length of the queue used could be optimized for either faster intrusion detection or lower risk of false rejection based on the use case.

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